

Amjad Althabteh

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C++ PORTFOLIO (github : AmjadAlthabteh)

Glint 3D Rendering Engine | C++17/OpenGL

[demo](#)

Glint-Rendering

- Designed and implemented a custom C++ memory pool allocator using placement new, aligned storage, and free-list reuse, reducing repeated heap allocations and allocation overhead by approximately 60%.
- Optimized rendering performance using frustum culling and GPU instanced rendering, cutting draw calls by ~40%.
- Improved startup performance by storing frequently accessed asset and simulation data in cache-friendly contiguous containers, reducing cold-start latency through sequential memory access and L1/L2 cache warming.

HFT Order Book | Low-Latency Performance System

HFT-lowlatency

- Built a C++20 low-latency order book using STL containers, move semantics, smart pointers, memory alignment, price-time priority, and unit-tested order matching for reliable trade execution.
- Reduced latency and memory contention using 64-byte cache-aligned (alignas(64)) structures, false-sharing prevention, and p50/p99/p99.9 latency benchmarks under concurrent load.

3D Space Simulation Engine

[demo](#)

OrbitalCollision

- Built a modular C++17/OpenGL orbital collision prediction engine that modeled 50+ orbital bodies using Newtonian gravity, vector mathematics, and RK4 numerical integration across more than 1M trajectory calculations.
- Reduced simulation runtime by approximately 40% using spatial partitioning, multithreaded trajectory propagation, SIMD-friendly math, branch reduction, function inlining, and -O3 compiler optimizations.
- Leveraged OpenGL, SFML, and ImGui to implement real-time rendering, visualization, and debugging tools for the simulation.

IronPath Robotic Control Systems

Robotic-ironpath

- Built a multithreaded C++ robotic control system using std::chrono, atomics, and CSV telemetry, sustaining a 100 Hz control loop with 142 µs average latency and 540 µs p99 latency.
- Created a custom memory leak detector by overloading new/delete, improving memory diagnostics and application reliability.

EXPERIENCE

Immersalab Graphics | Detroit, MI

05/25 – 02/26

Graphics Engineer

- Built internal C++/JSON tools and optimized GLSL shader compilation and asset-processing pipelines to improve rendering workflow efficiency.
- Reduced GPU idle time by approximately 30% through RenderDoc profiling and pipeline optimizations, improving frame time, submission latency, throughput, and GPU utilization.

Technologies Used: C++, OpenGL, GLSL, Python, JSON

UE Game Programmer C++ | Yorktown, VA

04/26 – 08/26

Grand Forge Games

- Built gameplay abilities in C++ using Unreal Engine's Gameplay Ability System (GAS) with Blueprint integration for game mechanics.
- Improved animation performance using layered blending and reusable C++ gameplay systems.

Technologies Used: C++, Unreal Engine, Blueprints visual scripting, JavaScript

Rare Munchies | Ann Arbor, MI

09/23 – 04/24

Software Development Intern

- Reduced unnecessary re-renders by ~35% through improved state management and component optimization.
- Built reusable TypeScript component systems focused on runtime efficiency and minimal overhead.

Wayne State University | Detroit, MI

02/25 – 12/25

Undergraduate Researcher – Spacecraft Dynamics & Collision Analysis

- Researched orbital trajectories and spacecraft collision risks across 10,000+ space objects using mathematical prediction models.
- Improved collision-analysis performance by approximately 30% using spatial partitioning, enabling faster large-scale simulations.

SKILLS

Languages: C++ (11–23), Rust, C, C#, Python, R, Assembly, q/KDB+, Scripting (Bash/PowerShell), Docker

Graphics & Engines: OpenGL, GLSL, Unreal Engine, Unity, Godot, Vulkan

Systems: Performance optimization, multithreading, lock-free data structures, memory pools, cache optimization, profiling, Linux, GDB, LLDB

EDUCATION

Wayne State University

B.S. in Computer Science

Detroit, MI

(top 6% in class) GPA: 3.77